

Homework 04 - Sampling & reconstruction

Deadline: 2nd test (date to be determined but not earlier than the 9th of April)

The mandatory questions account for 100 points, while the optional questions together contribute up to an additional 50 points.

1. Prove that for $x(t) = \cos(2\pi ft)$ and $x'(t) = \cos(2\pi f't)$ where $f' = f + kf_s, k \in \mathbb{Z}$, if sampled with frequency $f_s \rightarrow x[n] = x'[n]$.
2. Write a short script demonstrating the effect of aliasing similarly to the figures from the “Aliasing” slide in `dsp04.pdf`.
3. Starting from the definitions:

$$X(f) \equiv \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t)e^{2\pi i f t} dt, \quad \Pi_T(t) = T \cdot \sum_{n=-\infty}^{\infty} \delta(t-nT), \quad \hat{x}(t) = x(t) \cdot \Pi_T(t), \quad x[n] = x(nT).$$

$$\text{rect}(t) = \begin{cases} 1, & |t| \leq \frac{1}{2} \\ 0, & |t| > \frac{1}{2} \end{cases}, \quad \text{sinc}_{\pi}(t) = \frac{\sin(\pi t)}{\pi t}$$

Show that

(a)

$$\int_{-\infty}^{\infty} e^{2\pi i f t} df = \delta(t)$$

(g) 

$$\mathcal{F}[\text{rect}(t/T)] = T \text{sinc}_{\pi}(fT)$$

(b)

$$\mathcal{F}[\delta(t - \tau)] = e^{2\pi i f \tau}$$

(h) $\text{sinc}_{\pi}(m - n) = \delta_{mn}, m, n \in \mathbb{Z}$

(c)

$$\mathcal{F}[\text{rect}(t)] = \text{sinc}_{\pi}(f)$$

(i)

$$\hat{X}(f) \equiv \mathcal{F}[\hat{x}(t)](f) = \sum_{n=-\infty}^{\infty} X(f - nf_s)$$

(d) 

$$\int_0^{\infty} \frac{\sin \alpha t}{t} dt = \frac{\pi}{2} \text{sign}(\alpha).$$

(j)

$$\tilde{x}(t) \equiv \mathcal{F}^{-1}[\hat{X}(f) \cdot \text{rect}(f/f_s)](t) =$$

Hint: Make use of $d/d\alpha \left[\int_0^{\infty} \sin(\alpha t)/t dt \right] = \int_0^{\infty} \cos(\alpha t) dt$

(e) 

$$\mathcal{F}^{-1}[\text{sinc}_{\pi}(t)] = \text{rect}(f)$$

$$= \sum_{n=-\infty}^{\infty} x[n] \frac{\sin \pi(t - nT)}{\pi(t - nT)}$$

(f) 

$$\int_{-\infty}^{\infty} \text{rect}(t) dt = \int_{-\infty}^{\infty} \text{sinc}_{\pi}(t) dt = 1$$

(k) Argue that $\tilde{x}(t)$ is identical with $x(t)$ if the latter is bandwidth limited to the $[-f_s/2, f_s/2]$ interval.

4. Using the `fft` and `ifft` functions from Python demonstrate the validity of the following relations (plot on top of each other the result of the numerical Fourier transform and the corresponding analytical expression):

(a)

$$\mathcal{F}[e^{-t^2}] = \sqrt{\pi} e^{-\pi^2 f^2}$$

(d)

$$\mathcal{F}\left[\frac{1}{1+t^2}\right] = \pi e^{-2\pi|f|}$$

(b)

$$\mathcal{F}[\text{rect}(t)] = \text{sinc}_{\pi}(f)$$

Use log-normal scale and explain the anomaly!

(e)

(c)

$$\mathcal{F}[\text{sinc}_{\pi}(t)] = \text{rect}(f)$$

$$\mathcal{F}[e^{-|t|}] = \frac{2}{1 + 4\pi^2 f^2}$$

Explain the numeric anomaly!

Use log-log scale, compare the level of numeric anomaly to the previous case and explain it!